kmerseek, 152 alphabet-ksize settings (19 alphabets), notebook 241's search. Target database: 19_732 GENCODE v49 canonical human proteins. Query Ced9, target the BCL-2 family

Each block: Ced9 (top), identical residues (|), BCL-2 family member (bottom); 1-based coordinates. Bold name: call in register on the member's BH1 (UniProt).

BAK1: dayhoff6 k = 11, E = 23.9 (whole database), p = 0.0004 (this protein), 6 of 25 identical

```
Ced9     163 QCPMSYGRLIGLISFGGFVAAKMME 187
                 ||   |  ||   |
BAK1     120 ESGINWGRVVALLGFGYRLALHVYQ 144
```

BAX: dayhoff6 k = 7, E = 12224.9 (whole database), p = 0.1869 (this protein), 5 of 21 identical

```
Ced9     168 YGRLIGLISFGGFVAAKMMES 188
              ||   |  |       |
BAX      107 WGRVVALFYFASKLVLKALCT 127
```

BCL2: hp_lehninger2 k = 19, E = 11172.7 (whole database), p = 0.2097 (this protein), 5 of 26 identical

```
Ced9     163 QCPMSYGRLIGLISFGGFVAAKMMES 188
                 ||       |||
BCL2     139 RDGVNWGRIVAFFEFGGVMCVESVNR 164
```

BCL2A1: dayhoff6 k = 7, E = 185.6 (whole database), p = 0.0029 (this protein), 6 of 31 identical

```
Ced9     156 VGNAQTDQCPMSYGRLIGLISFGGFVAAKMM 186
             |            ||      | |    |
BCL2A1    74 VMEKEFEDGIINWGRIVTIFAFEGILIKKLL 104
```

MCL1: dayhoff6 k = 12, E = 16.7 (whole database), p = 0.0005 (this protein), 10 of 18 identical

```
Ced9     168 YGRLIGLISFGGFVAAKM 185
              ||   ||||| |||
MCL1     261 WGRIVTLISFGAFVAKHL 278
```

BCL2L1: hp_thomas_dill_no_c2 k = 17, E = 479.9 (whole database), p = 0.0098 (this protein), 3 of 26 identical

```
Ced9     156 VGNAQTDQCPMSYGRLIGLISFGGFV 181
             |  |            |
BCL2L1    65 VNGATGHSSSLDAREVIPMAAVKQAL 90
```

BCL2L10: dayhoff6 k = 7, E = 7315.6 (whole database), p = 0.1232 (this protein), 5 of 17 identical

```
Ced9     160 QTDQCPMSYGRLIGLIS 176
             |         ||  | |
BCL2L10  130 QEGDVARDCQRLVALLS 146
```

BCL2L2: hp_pbotc_1st_ed2 k = 16, E = 2108.5 (whole database), p = 0.0352 (this protein), 4 of 58 identical

```
Ced9     129 ENFETFCEQLLAVPRISFSLYQDVVRTVGNAQTDQCPMSYGRLIGLISFGGFVAAKMM 186
                                                    | |    |  |
BCL2L2   132 TQLADWIHSSGGWAEFTALYGDGALEEARRLREGNWASVRTVLTGAVALGALVTVGAF 189
```

BOK: no call covers Ced9 residue 168.